ML-Based Spam Email Detection

1. Abstract

Spam emails are unwanted messages sent in bulk for advertising, phishing, fraud, or malicious purposes. These emails waste users' time and may compromise security. The objective of this project is to develop a Machine Learning-based Spam Email Detection System that can automatically classify incoming messages as either Spam or Ham (Not Spam).

The project uses the SMS Spam Collection Dataset from Kaggle and applies Natural Language Processing (NLP) techniques along with machine learning algorithms to analyze message content and predict whether a message is spam or legitimate.

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2. Introduction

Email and SMS communication are widely used around the world. Unfortunately, many users receive spam messages that contain advertisements, scams, or harmful links. Manually filtering such messages is inefficient and time-consuming.

Machine Learning provides an intelligent solution by learning patterns from historical data and automatically classifying new messages.

This project focuses on building a spam detection model using text classification techniques and machine learning algorithms.

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3. Problem Statement

Spam messages create security risks and reduce communication efficiency. The goal of this project is to develop a machine learning model capable of automatically detecting and filtering spam messages with high accuracy.

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4. Objectives

The main objectives of this project are:

- To understand the process of text classification.

- To preprocess textual data using NLP techniques.
- To convert text into numerical features.
- To train machine learning models for spam detection.
- To evaluate model performance using standard metrics.
- To classify messages as Spam or Ham.

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5. Dataset Description

Dataset Name: SMS Spam Collection Dataset

Source:
https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset

Dataset Characteristics:

- Total Messages: 5,572
- Features: 2
- Classes:
  - Spam
  - Ham (Not Spam)

Dataset Columns:

Column	Description
v1	Label (Spam/Ham)
v2	Message Text

Example:

Label	Message
Ham	I'm going home now
Spam	Congratulations! You've won a prize

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6. Machine Learning Workflow

The project follows the standard machine learning pipeline:

1. Data Collection
2. Data Preprocessing
3. Feature Extraction

4. Dataset Splitting
5. Model Training
6. Model Evaluation
7. Prediction

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7. Data Preprocessing

Data preprocessing is performed to clean and prepare the text data before training.

Steps Performed

1. Lowercase Conversion

All messages are converted to lowercase.

Example:

Before:
WIN A FREE PRIZE

After:
win a free prize

2. Removal of Punctuation

Special characters such as:

.,!?@#$%^&*

are removed.

3. Tokenization

Messages are split into individual words.

Example:

"Win free cash now"

becomes

["win", "free", "cash", "now"]

4. Stop Word Removal

Common words that do not contribute much meaning are removed.

Examples:

- is
- am
- the
- and
- a

5. Stemming

Words are reduced to their root form.

Examples:

running → run

playing → play

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8. Feature Extraction

Machine learning algorithms cannot directly understand text.

Therefore, text messages are converted into numerical vectors using:

TF-IDF Vectorization

TF-IDF stands for:

Term Frequency – Inverse Document Frequency

It measures the importance of words in a message.

Advantages:

- Captures important keywords.
- Reduces impact of common words.
- Improves classification performance.

9. Dataset Splitting

The dataset is divided into:

Dataset	Percentage
Training Data	80%
Testing Data	20%

Training data is used to learn patterns.

Testing data is used to evaluate performance.

10. Machine Learning Algorithm

Multinomial Naive Bayes

The primary algorithm used is Multinomial Naive Bayes.

Reasons for Selection:

- Works well with text data.
- Fast training and prediction.
- High performance on spam detection tasks.
- Simple implementation.

Working Principle:

The algorithm calculates the probability that a message belongs to the Spam or Ham class and assigns the class with the highest probability.

11. Model Training

The model is trained using:

- Preprocessed text messages
- TF-IDF feature vectors

- Training dataset

The model learns common patterns found in spam and legitimate messages.

Examples:

Spam keywords:

- win
- free
- prize
- cash
- claim

Ham keywords:

- meeting
- home
- dinner
- call
- office

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12. Model Evaluation

The trained model is evaluated using the testing dataset.

Evaluation Metrics

Accuracy

Measures overall correctness.

Accuracy = Correct Predictions / Total Predictions

Precision

Measures how many predicted spam messages were actually spam.

Recall

Measures how many actual spam messages were successfully detected.

F1-Score

Harmonic mean of Precision and Recall.

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13. Confusion Matrix

A confusion matrix summarizes prediction results.

	Predicted Ham	**Predicted Spam**
Actual Ham	TN	FP
Actual Spam	FN	TP

Where:

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

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14. Results

After training and testing, the model achieved high performance in spam classification.

Typical Results:

- Accuracy: 97% – 99%
- Precision: High
- Recall: High
- F1-Score: High

The model successfully identifies most spam messages while minimizing false alarms.

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15. Advantages

- Automatic spam filtering.
- Fast prediction.
- Improves communication security.
- Reduces manual effort.
- Easy to deploy.

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16. Limitations

- Performance depends on training data quality.
- New spam patterns may reduce accuracy.
- Requires retraining with updated datasets.
- Cannot fully understand message context.

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17. Applications

- Email Spam Detection
- SMS Filtering
- Fraud Detection
- Phishing Detection
- Customer Support Systems
- Social Media Message Filtering

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18. Future Enhancements

Future improvements may include:

- Deep Learning models.
- LSTM Networks.
- BERT-based text classification.
- Real-time spam filtering systems.
- Multi-language spam detection.
- Web and mobile deployment.

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19. Conclusion

This project successfully demonstrates the use of Machine Learning and Natural Language Processing for spam detection. The SMS Spam Collection Dataset was preprocessed and transformed into numerical features using TF-IDF. A Multinomial Naive Bayes classifier was trained and evaluated to classify messages as Spam or Ham. The model achieved high accuracy and proved effective for text classification tasks. This project provides a strong foundation for understanding beginner-level machine learning workflows and real-world applications of NLP.